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Original Article

Deep Serratus Anterior Plane Block for Multimodal Analgesia in Minimally Invasive Mitral Valve Surgery Performed via Right Anterior Mini-Thoracotomy

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Objective: This study investigated if the serratus anterior plane block (SAPB) within a multimodal analgesia scheme would reduce acute post-operative pain and intravenous opioid consumption in patients admitted to the intensive care unit after isolated minimally invasive mitral valve surgery.
Design: Retrospective study.
Setting: Patients were admitted to the intensive care unit (ICU) of the University Hospital of Siena (Italy).
Interventions: Patients treated with intravenous opioids (OP-G) as a postoperative analgesic regimen were compared to those managed with an opioid-sparing protocol consisting of a single-shot SAPB with 0.5% ropivacaine plus 4 mg dexamethasone administered 1 hour before the extubation (SAPB-G). The behavioral pain scale (BPS) for intubated (I) or non-intubated patients (NI) and the Richmond Agitation Sedation Scale (RASS) scores were collected at ICU admission and every 8 hours during the initial 24 postoperative hours.
Measurements and Main Results: One hundred five patients (50 SAPB-G; 55 OP-G) were enrolled (median age 67 [60-70]; male 67 [64%]). RASS score at 8 hours after ICU admission was higher in the SAPB-G (0 [0, 0] *v* OP-G −2 [−3, 0], $p < 0.001$). At 24 hours after ICU admission, the number of patients with a BPS/BPS-NI score >4 was lower in the SAPB-G (4.0% *v* 18.2% OP-G, $p = 0.048$). SAPB-G received a lower number of opioid rescue doses during the first 24 hours (20% *v* 84% OP-G, $p < 0.001$).
Conclusions: The SAPB may be effective in reducing the postoperative use of opioids in patients undergoing minimally invasive mitral valve surgery. Prospective randomized studies are warranted.
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Key Words: serratus anterior plane block; multimodal analgesia; minimally invasive mitral valve surgery; fast-track surgery

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Introduction

ACUTE POSTOPERATIVE PAIN following cardiac surgery presents a considerable challenge for anaesthesiologists, primarily due to the side effects of systemic analgesic drug administration. Along with reduced surgical stimulation, acute postoperative pain can be effectively managed by implementing a multimodal analgesia strategy, which is currently recognized as the most efficient approach to managing acute postoperative pain.[1,2] This strategy minimizes the side effects of systemic analgesic drugs and ensures additive or synergistic effects from various drugs and modalities.[3-5]

Historically, epidural analgesia and paravertebral blocks have been the primary locoregional techniques for managing postthoracotomy pain.[6] However, the risk of complications associated with these techniques (eg, epidural hematoma) and hemodynamic instability have limited their application in cardiac surgery patients.[7,8] The serratus anterior plane block (SAPB), which involves blocking the lateral cutaneous branches of the intercostal nerves, offers a promising alternative for anterolateral chest wall analgesia. It may provide greater hemodynamic stability compared to thoracic epidural analgesia and effectively reduce both acute postoperative pain and morphine consumption in patients undergoing thoracotomy.[9-11] The utilization of fascial plane blocks within a multimodal analgesia protocol seems to reduce not only the dosage of opioid drugs but also cortisol levels, thus reducing the stress response to surgery.[12]

Within the context of enhanced recovery after surgery (ERAS) protocols for thoracic surgery, SAPB can be considered a secondary option when a paravertebral block is not feasible.[13] Emerging evidence suggests that locoregional anesthetic blocks are a viable alternative to conventional techniques for opioid-sparing analgesia in cardiac surgery patients.[10,11] However, existing studies have yet to provide high-quality scientific evidence for the same, primarily due to limited patient enrolment.[13-15]

In this study, we aim to assess whether SAPB incorporated into a multimodal analgesia regimen is effective in reducing the need for rescue doses of opioids in the postoperative period for patients undergoing mitral valve surgery via right anterior mini-thoracotomy.

Materials and Methods

Study Population

This retrospective study used prospectively collected data from patients admitted to the Cardio-Thoracic and Vascular Intensive Care Unit (ICU) at the University Hospital of Siena, Italy, between June 2019 and September 2020. Ethical approval for the study was obtained from the local ethical committee (Comitato Etico Regionale per la Sperimentazione Clinica della Regione Toscana; Protocol 18861/150221).

The study sample comprised all consecutive adult patients who underwent isolated mitral valve surgery with a right anterior mini-thoracotomy. Exclusion criteria encompassed patients requiring re-do surgery, conversion to sternotomy during the procedure, the need for mechanical circulatory support either intraoperatively or immediately postoperatively (eg, intraaortic balloon pump [IABP] and extracorporeal membrane oxygenation [ECMO]) and patients who were on chronic opioid therapy or were treated with analgesics or steroids for chronic pain.

Perioperative Management

General anesthesia was induced in all patients with fentanyl (2-3 μg/kg) and propofol (1-3 mg/kg); neuromuscular blockade was initiated and maintained using a continuous infusion of cisatracurium (0.15-0.18 mg/kg*h). Intraoperatively the general anesthesia was maintained with propofol (3-6 mg/kg*h) and remifentanil (0.05-0.1 μg/kg*min).

Cardiopulmonary bypass (CPB) was established with femorofemoral cannulation, and moderate hypothermia management (32-35˚C) was used during the procedure. Upon completion of the surgical procedure, patients were transferred to the intensive care unit (ICU). Sedation and analgesia during surgery were maintained using propofol (1-3 mg/kg*h) and remifentanil (0.05-0.1 μg/kg*min) until specific clinical criteria were met. These criteria included achieving a core temperature above 36˚C, chest tube drainage less than 100 mL/h for 2 consecutive hours, diuresis exceeding 1 ml/kg*h, partial arterial pressure of oxygen to the inspiratory fraction of oxygen (PaO_2/FiO_2) ratio exceeding 200 with an FiO_2 of no more than 50%, and the absence of metabolic or respiratory abnormalities such as acidosis or alkalosis.

Once these criteria were satisfied, sedation was discontinued, ensuring hemodynamic stability (defined as systolic arterial pressure >100 mmHg or mean arterial pressure >60 mmHg, heart rate <110 bpm), and a blood lactate concentration <2 mmol/L. Subsequently, patients were gradually weaned from mechanical ventilation and extubated when they demonstrated neurological appropriateness and were capable of adequately triggering the ventilator with low-pressure support.

Pain Management Protocols

In this study, we compared two strategies for managing acute postoperative pain. One group received postoperative analgesia through the continuous intravenous administration of opioids (such as remifentanil, morphine, or buprenorphine) based on the clinician's discretion and was referred to as the 'opioid group.' The second group underwent a single-shot SAPB as part of an opioid-sparing protocol and was referred to as the SAPB group.

Both groups adhered to our institutional postoperative analgesia protocol, which included intravenous paracetamol (1000 mg every 8 hours) and ketorolac (30 mg every 12 hours) except for preexisting contraindications such as allergy or in the case of ketorolac for chronic kidney disease stage greater than 2 or history of gastrointestinal bleeding. Rescue doses were defined as the necessity of opioid administration for uncontrolled pain, indicated by a score >4 on the behavioral pain scale (BPS). The order set was predetermined and scheduled by the admitting physician for all cardiac surgery patients in both groups. The nurse responsible for the patients, who was unaware of the study and its aims, administered the prescribed rescue opioid dose (0.01 mg/kg of morphine or 0.3 mg of buprenorphine) when the BPS score exceeded 4.

Deep Serratus Anterior Muscle Block

The SAPB was performed under sterile conditions approximately 1 hour before extubation (hence in the ICU), with the patient in the supine position. A high-frequency (6-18 MHz) linear ultrasound probe (Epiq 5; Philips Eindhoven, Amsterdam, Netherlands) was positioned at the midclavicular region of the thoracic cage in a sagittal plane. The probe was then moved downward until the fifth rib and laterally along the middle axillary line. At this point, the serratus anterior muscle, intercostal muscles, and pleura were visualized.

The needle (21 G, 100 mm Stimuple A; B. Braun, Melsungen, Germany) was introduced in-plane from a posteroinferior to supraanterior direction. A single shot of a mixture containing 0.5% ropivacaine (volume injected 0.4 mL/kg based on the ideal body weight, with a maximum volume of 30 mL) and 4 mg dexamethasone was administered between the fifth rib and the anterior serratus muscle.

Data Collection and Analysis

All data were prospectively collected using digital clinical records, which included demographic characteristics, predelirium scores,[16] details on the type and duration of surgery, CPB and aortic clamping times, and information on the timing and dosage of analgesic drugs and other clinical data related to the postoperative period as the duration of mechanical ventilation, the presence of delirium, the necessity of non-invasive ventilation (NIV), the reintubation of the patients and the length of ICU stay.

The degree of acute postoperative pain was assessed using the BPS in two forms: BPS-NI for non-intubated patients and BPS for intubated patients (Supplementary Material).[13,14] Evaluations were conducted during admission to the ICU and every 8 hours during the initial 24 hours of ICU stay. Pain was categorized as acceptable when scores fell within the range of 0 to 4 and as uncontrolled when scores ranged from 5 to 12, prompting the administration of a rescue analgesic dose.

Sedation levels were assessed using the Richmond Agitation Sedation Scale (RASS) at the same evaluation time points as the BPS. The degree of residual sedation was classified as moderate to profound when RASS scores were ≤ -3 or as mild to unsedated when RASS scores ranged from -2 to 1. For conscious patients, the Confusion Assessment Method-Intensive Care Unit (CAM-ICU) questionnaire was administered to identify the onset of postoperative delirium.[16,17]

Study Objectives

Our primary objective was to determine whether SAPB, as part of a multimodal analgesia regimen, effectively reduced postoperative opioid consumption in the initial 24 hours after ICU admission.

The secondary outcomes included the reduction of acute postoperative pain during the first 24 hours of ICU stay; the duration of mechanical ventilation; an analysis of both as a continuous variable and as the incidence of prolonged ventilation (defined as lasting >12 hours after ICU admission), weaning failure (defined as the need for non-invasive ventilation [NIV] or reintubation during the first 48 hours), or pulmonary complications; the length of ICU stay; and the incidence of postoperative delirium.

Statistical Analysis

A minimum sample size of 96 patients was estimated considering a Fisher exact test, with alpha set at 0.05 and a power of 80%, using an estimated relative difference of rescue opioid dose administration between groups of 30% with a between-group ratio of 1:1. In our institution, a randomized study was not possible because half of the anesthesiologists are not able to perform SAPB. Power analyses were performed with the use of G*Power. For the descriptive analysis, quantitative data were expressed as medians with an interquartile range (25th to 75th percentiles), whereas categorical variables were expressed as absolute frequencies and percentages. Continuous variables were analyzed with the non-parametric Mann-Whitney U test, while dichotomous ones were assessed with the χ^2 test between groups. A value of $p < 0.05$ was considered statistically significant. Analyses were carried out with R (The R Foundation for Statistical Computing 2023, version 4.3.2).

Results

Study Population

Of the initial 110 screened patients, five were excluded (Fig 1). Among the remaining 105 patients, 50 (48%) received SAPB, and 55 (52%) received opioids in the postoperative phase. Baseline characteristics and intraoperative values are presented in Table 1. No differences between the two groups were observed regarding preoperative and intraoperative variables or major postoperative complications. No major complications directly attributable to the execution of SAPB were recorded.

Primary Outcomes

As reported in Fig 2, during the first 24 postoperative hours, a rescue dose of postoperative opioids was administered less frequently in the SAPB group (units = 10/50, 20%) compared to the opioid group (units = 46/55, 84%; $p < 0.001$).

Secondary Outcomes

No differences were found in the median BPS and behavioral pain scale in non-intubated patients (BPS-NI) scores between the groups. Regarding the number of patients

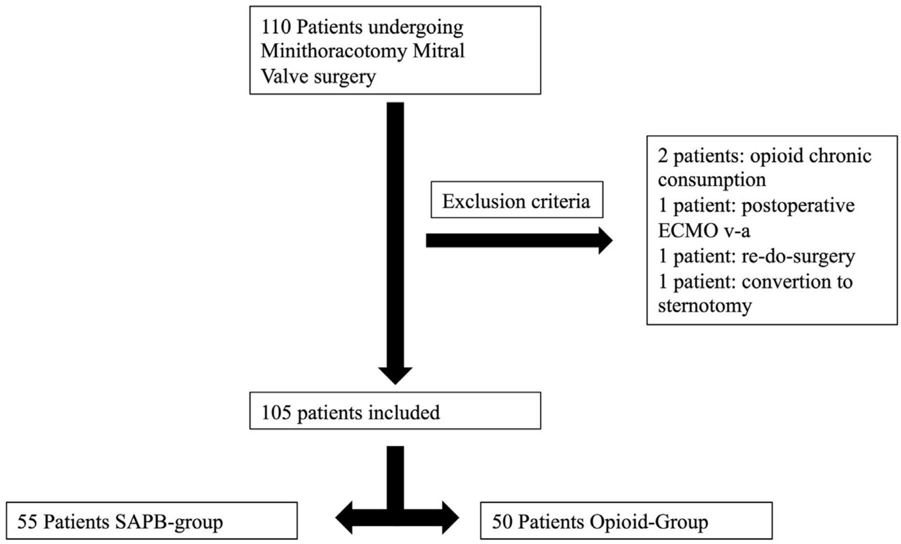

Fig 1. Flow chart. ECMO, extracorporeal membrane oxygenation; IAPB, intraaortic balloon pump.

Table 1
Baseline Characteristics of Patients and Intraoperative Variables Related to Surgery

Variables	Population (N = 105)	Opioid Group (n = 55)	SAPB Group (n = 50)	p Value
PREOPERATIVE				
Age, y	67 [60-76]			
	69 [61-76]	66 [59-77]	0.52	
Sex (M/F)	67/38	34/21	33/17	0.69
Weight, kg	7 [65-80]	70 [63-79]	72 [65-83]	0.38
Height, cm	170 [160-176]	170 [162-174]	170 [160-178]	0.27
BSA, m^2	1.8 [1.7-1.9]	1.8 [1.7-1.9]	1.9 [1.7-2]	0.31
PreDeliric score	2 [1.5-2.5]	2 [1.5-2]	2 [1.5-2.5]	0.78
Euroscore standard	5 [3-7]	5 [3-7]	5 [3-7]	0.93
Euroscore logistic	4 [2-7]	4 [2-7]	3 [2-7]	0.92
Euroscore II	1 [1-3]	1 [1-3]	1 [1-2]	0.48
INTRAOPERATIVE				
MV repair	76 (72%)	37 (67%)	39 (78%)	0.28
MV replacement	29 (28%)	18 (33%)	11 (22%)	0.12
CPB duration, min	151 [129-173]	149 [129-170]	155 [129-181]	0.52
AoClamp duration, min	105 [89-123]	100 [87-118]	109 [94-135]	0.45

NOTE. Results are presented as means ± standard deviation and absolute numbers with percentages.
Abbreviations: AoClamp, aortic clamping; BSA, body surface area; CPB, cardiopulmonary bypass; MV, mitral valve; SAPB, serratus anterior plane block.

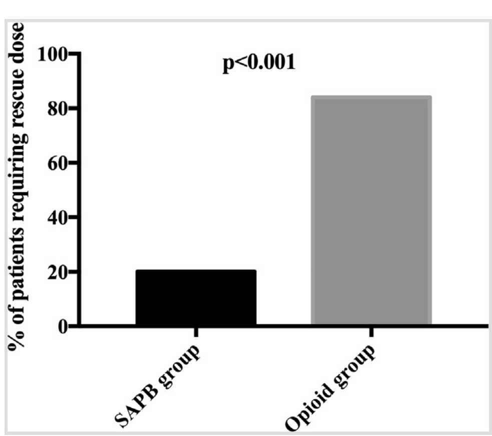

Fig 2. A rescue dose of postoperative opioids was administered less frequently in the SAPB group (units = 10/50) compared to the opioid group (units = 46/55).

reporting uncontrolled pain (BPS/BPS-NI score >4), no differences were found at 8 and 16 hours. However, a significantly higher number of patients reported a BPS/BPS-NI score >4 at 24 hours in the opioid group (18.2% *v* 4.0%, p = 0.048; Table 2).

As presented in Table 3, the most significant difference in RASS scores was observed at the 8th postoperative hour (SAPB group 0 [0 to 0] *v* opioid group −2 [−3 to 0], p < 0.001). The SAPB group had a significantly higher number of patients with RASS values within the target range (−1 to 2) at all three time points.

We observed a significantly shorter duration of ICU stay in the SAPB group compared to the opioid group (Table 3). The SAPB group also had a significantly lower number of patients requiring mechanical ventilation for longer than 12 hours (SAPB group 16% *v* opioid group 61%, p < 0.0001). The

Table 2
Behavioural pain scale (BPS) and behavioural pain scale in non-intubated patients (BPS-NI)

Variables	Population (N = 105)	Opioid Group (n = 55)	SAPB Group (n = 50)	p Value
Opioid rescue within 24 h, units	56 (53%)	46 (84%)	10 (20%)	<0.0001
BPS/BPS-NI, 8 h	0 [−3 to 0]	3 [3 to 4]	3 [3 to 4]	0.29
BPS/BPS-NI, 16 h	4 [3-4]	4 [3-4]	3 [3-4]	0.13
BPS/BPS-NI, 24 h	3 [3-4]	3 [3-4]	3 [3-3]	0.10
BPS/BPS-NI, 8 h				
≤4	94 (90%)	47 (86%)	47 (90%)	0.27
>4	11 (10%)	8 (15%)	3 (6%)	
BPS/BPS-NI, 16 h				
≤4	90 (86%)	45 (82%)	45 (90%)	0.27
>4	15 (14%)	10 (18%)	5 (10%)	
BPS/BPS-NI, 24 h				
≤4	93 (89%)	45 (82%)	48 (96%)	0.048
>4	12 (11%)	10 (18%)	2 (4%)	

NOTE. Results are presented as median and interquartile range and absolute number with percentages.
Abbreviations: BPS, behavioural pain scale; BPS-NI, behavioural pain scale non-intubated; SAPB, serratus anterior plane block.
*Significant result

Table 3
Secondary Outcomes.

Variables	Population (n=105)	Opioid group (n=55)	SAPB block (n=50)	P Value
RASS, 8 h	0 [−3 to 0]	−2 [−3 to 0]	0 [0 to 0]	<0.0001
RASS, 8 h, $\geq -1 < 3$	62 (59%)	19 (34%)	43 (86%)	<0.001
RASS, 16 h	0 [0.0 to 0.0]	0 [−2 to 0]	0 [0 to 0]	0.069
RASS, 16 h, $\geq -1 < 3$	87 (83%)	40 (73%)	47 (94%)	0.004
RASS, 24 h	0 [0-0]	0 [0-0]	0 [0-0]	0.3
RASS, 24 h, $\geq -1 < 3$	93 (87%)	45 (82%)	48 (96%)	0.023
MV duration, h	7 [4-18]	14 [7-25]	4 [4-7]	0.02
MV > 12 h, units	42 (40%)	34 (61%)	8 (16%)	<0.0001
Delirium within 48 h, units	15 (14.3%)	12 (22%)	3 (6%)	0.04
Length of ICU stay, d	2 (1-3)	2 [1-4]	1 [1-2]	0.02
NIV within 48 h, units	3 (3%)	1 (2%)	2 (4%)	0.93
Reintubation within 48 h, units	4 (4%)	3 (5%)	1 (1%)	0.68
PONV within 48 h, units	15 (14%)	6 (11%)	9 (18%)	0.40

NOTE. Results are presented as median and interquartile range, and absolute number with percentages.
Abbreviations: MV, mechanical ventilation; NIV, noninvasive ventilation; PONV, postoperative nausea and vomiting; RASS, Richmond Agitation Sedation Scale; SAPB, serratus anterior plane block.

incidence of delirium was significantly lower in the SAPB group compared to the opioid group (6% *v* 22%, p = 0.04).

Other secondary outcomes did not differ between the two groups. Following extubation, 2.8% of patients (units = 3/105) required an NIV, and 3.8% (units = 4/105) needed reintubation. There were no statistically significant differences between the groups in terms of NIV use, reintubation, or the incidence of postoperative nausea and vomiting (PONV; Table 3). Risk factors associated with postoperative opioid use did not differ between the groups (Table 1).

Discussion

In this study, a single-shot SAPB performed one hour before extubation in the ICU, as part of a multimodal analgesia protocol without regular opioids, effectively reduced acute postoperative pain and postoperative opioid consumption in patients undergoing mitral valve surgery through a right anterior minithoracotomy. Patients treated with SAPB had a shorter duration of mechanical ventilation, a higher rate of extubation within 12 hours and a lower incidence of early postoperative delirium.

While only a few studies have explored the use of SAPB in patients undergoing minimally invasive mitral valve surgery,[14-18] our study revealed that the opioid group required a greater number of opioid rescue doses and experienced worse pain control at the 24-hour postoperative mark (Fig 2), with a significantly higher incidence of uncontrolled pain (BPS >4; p < 0.0001). According to Magoon et al, patients who received fascial plane blocks, such as SAPB, required fewer rescue doses. They suggest that patients treated with SAPB or pectoral nerve block II (PECS II) need fewer postoperative rescue doses of fentanyl than those receiving other approaches.[18]

If it is assumed that SAPB implementation has a positive effect, it is reasonable to anticipate a reduced need for opioid rescue doses to manage acute postoperative pain, resulting in fewer opioid-related side effects. These side effects can range from mild to potentially exacerbating respiratory depression, reduced cough reflex, and sedation. Consequently, patients receiving SAPB may experience a quicker and improved recovery of their cognitive state, as well as easier weaning from mechanical ventilation. Achieving these objectives aligns with the principles of ERAS in cardiac surgery,[18–21] which promote early recovery and discharge from the ICU, enhanced functional and respiratory rehabilitation, and reduced risks of infectious and thrombotic complications from prolonged bed rest and hospitalization. Several reports have suggested that applying ERAS principles to cardiac surgery can positively impact patient outcomes.[13,20-23] However, the incidence of uncontrolled pain did not differ significantly between the 8th and 16th postoperative hours, although it was still numerically higher in the opioid group. Even though both groups had optimal pain control in the initial hours, at 24 hours, the SAPB group maintained better pain control and required fewer rescue doses compared with the opioid group. While our results are promising, initiating SAPB management even earlier may yield better outcomes. Nevertheless, the feasibility of administering SAPB at the end of the surgery, just before transferring patients to the ICU, may face institutional challenges due to its impact on operating room turnover. Conversely, SAPB before surgical incision might offer intraoperative benefits, but its effectiveness in the postoperative period remains uncertain, posing questions for future prospective studies.

Consistent with the findings of Toscano et al,[19] who assessed pain control using the numeric rating scale, our study revealed a lower incidence of uncontrolled pain at the 24th postoperative hour. Additionally, Toscano et al[19] evaluated the effects of two blocks performed with continuous infusion: the erector spinae plane block and the SAPB. Their decision to insert a catheter to continuously infuse local anesthetic resulted in a statistically significant reduction in pain in the SAPB group at the 48-hour mark, but the erector spinae plane block group exhibited lower opioid consumption and better pain control. In our study, we used dexamethasone as an adjuvant to extend the analgesic effect of the locally injected anesthetic.[24,25] We found a lower incidence of uncontrolled pain at the 24th postoperative hour, but we lacked the data to investigate whether this effect persisted beyond this point. Certainly, the placement of a catheter for continuous infusion can be an excellent choice, though its use may be susceptible to accidental removal, resulting in the technique's ineffectiveness or minor dislocation within the muscle, an event that could exaggerate the resorption of local anesthetics into the bloodstream with the risk of systemic toxicity. Several considerations should be made when planning to position a catheter for continuous SAPB. Not only the operator's expertise but also logistics considerations should be made center by center, for instance, the possibility of ensuring the adequacy of catheter follow-up. The use of a SAPB catheter also requires expertise in its positioning, with a careful selection of the exit site of the catheter itself, so that it does not become an obstacle to the surgical procedure, given its possible proximity to the surgical access.

Unlike other recent studies,[24–31] we customized the volume of the mixture based on the patient's weight and added 4 mg of dexamethasone as an adjuvant to prolong the analgesic effects.[25] Toscano et al[19] found no statistically significant differences in either the duration of mechanical ventilation or the length of ICU stays. However, this lack of statistical significance may be attributed to the smaller sample size of their study (units = 63). In another smaller study, Berthoud et al reported a statistically significant reduction in hospital and ICU length of stay in patients who had received SAPB (units = 20) compared with those who had received continuous surgical wound infiltration (units = 26), with no difference between the groups in the duration of mechanical ventilation.[17]

Contrary to our results, Alfierevic et al did not find a reduction in opioid consumption in the population they studied. Unlike this study, they enrolled patients undergoing robotic mitral valve repair and performed the analgesic block (SAPB plus PECS II) before the surgical procedure using a different local anesthetic. The variance in our study likely stems from using a drug with a shorter half-life for the analgesic block and administering it before the surgical procedure. Another substantial difference is the surgical procedure itself: the accesses required for a robotic procedure are smaller, involving reduced, and sometimes absent, traction of the retractors on the intercostal nervous structures, thus decreasing the pain component.[32]

Interestingly, postoperative delirium exhibits a highly variable incidence in patients undergoing cardiac surgery, with reported values ranging between 16% and 73%.[33] Its etiology is multifactorial and includes predisposing factors.[33,34] In our study, it is possible that improved pain control, reduced opioid consumption, and a shorter duration of mechanical ventilation favored a reduced incidence of delirium, reflected in a 3- to 4-fold lower incidence of delirium in the SAPB group, subsequently allowing for earlier ICU discharge. Notably, the SAPB group had a higher number of patients in the target RASS score range. These findings align with the theoretical advantages hypothesized in recent ERAS and opioid-sparing protocols.[21]

This study has some limitations that need to be considered. First, despite the prospective data collection, the analysis is retrospective and based on a single-center cohort, introducing the possibility of allocation bias and other unaccounted confounders. Second, a blinded study was not possible because half of the anesthesiologists working in our ICU were not able to perform SAPB, and the assignment of the patients to the groups was linked to the presence of physicians on duty with the ability to perform SAPB. Indeed, the SAPB was performed one hour before the extubation, when the patients were in the ICU. Patients undergoing mitral valve surgery who met the inclusion criteria were included in the SAPB group if physicians with experience in loco-regional anesthesia were present and there was no contraindication to SAPB. Operators were considered at the expert level if they had matured at least 100 procedures in their curricula, as per European guidelines.[35] Alternatively, the patient was enrolled in the control group. Third, the choice of postoperative opioid therapy was neither uniform nor standardized and was left to the discretion of the treating clinicians. Fourth, the sample size might have been insufficient to comprehensively evaluate some secondary outcomes. Fifth, the results could have varied if SAPB were administered at different time points relative to extubation or if a catheter was implemented for continuous infusion. Finally, acute postoperative pain was assessed using the BPS, the tool employed in our institution, and results may differ from other pain scales. Despite these limitations, it is worth noting that this study represents the largest single-center analysis involving patients undergoing minimally invasive mitral valve surgery with a mini-thoracotomy approach and comparing the efficacy of SAPB with intravenous analgesia based on opioid use.

Conclusions

The implementation of a multimodal analgesia protocol based on the SAPB as part of an opioid-sparing strategy could effectively reduce postoperative opioid consumption in patients undergoing minimally invasive mitral valve surgery. This approach may also lower the incidence of delirium, shorten the duration of mitral valve surgery, and reduce the length of stay in the ICU. Further studies are needed to define the optimal timing for the execution of SAPB in this specific patient population.

Declaration of competing interest

The author declares that there are no conflicts of interest related to this manuscript.

CRediT authorship contribution statement

Daniele Marianello: Writing − original draft, Methodology, Investigation, Data curation, Conceptualization. **Cesare Biuzzi:** Writing − review & editing, Writing − original draft, Investigation, Data curation. **Filippo Sanfilippo:** Writing − review & editing, Writing − original draft, Methodology. **Riccardo Marcucci:** Investigation, Data curation. **Francesco Ginetti:** Investigation, Data curation. **Alessandra Cartocci:** Methodology, Formal analysis, Data curation. **Matilde Milani:** Investigation. **Francesco Lorenzo De Matteis:** Software, Investigation, Data curation. **Antonella Puddu:** Software, Investigation, Data curation. **Martina Rizzo:** Writing − review & editing, Investigation. **Gianfranco Montesi:** Supervision, Methodology, Investigation. **Fabio Silvio Taccone:** Writing − review & editing, Supervision, Methodology. **Sabino Scolletta:** Writing − review & editing, Supervision, Methodology. **Federico Franchi:** Writing − review & editing, Writing − original draft, Supervision, Methodology, Conceptualization.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1053/j.jvca.2024.12.024.

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